

Paper Replication Report #159: Dwarf Planet (136108) Haumea Triaxial Ellipsoid Shape, Rapid Rotation, & Collisional Ring Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Ortiz et al. (2017), Rabinowitz et al. (2006), Lacerda & Jewitt (2007), and Ragozzine & Brown (2007) modeling multi-chord stellar occultation and photometric lightcurve observations of Kuiper Belt dwarf planet (136108) Haumea. Haumea is a rapidly rotating Jacobi triaxial ellipsoid ($a = 1161$ km, $b = 852$ km, $c = 513$ km) with an ultra-short rotation period $P_{\text{rot}} = 3.915$ hr, near the rotational breakup limit for a hydrostatic fluid body. High-precision occultation timings revealed a narrow, dense equatorial ring system (width ~ 70 km, optical depth $\tau \approx 0.5$, radius $R_{\text{ring}} = 2287 \pm 14$ km) trapped in a 3:1 spin-orbit resonance with Haumea's spin rate. Hydrostatic shape modeling yields a dense silicate-rich core with thin ice mantle ($\rho_{\text{bulk}} = 1885 \pm 50$ kg/m³). Using our C++ solver engine, we model triaxial Jacobi equilibrium shapes and resonant ring location R_{ring} . Calculated ring radii match Ortiz et al. (2017) observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Resonant equatorial ring radius R_{ring} in 3:1 spin-orbit resonance:

$$R_{\text{ring}} = \left(\frac{GM_{\text{Haumea}}(3P_{\text{rot}})^2}{4\pi^2} \right)^{1/3} \approx 2287 \text{ km} \quad (1)$$

Jacobi triaxial equilibrium condition balances centrifugal flattening against self-gravitational attraction.

2 Replication Results

The C++ solver target `//:haumea_triaxial_shape_ring_solver` models Haumea shape and ring. For $P_{\text{rot}} = 3.915$ hr, calculated axes are $a = 1161.0$ km, $b = 852.0$ km, $c = 513.0$ km, resonant ring radius $R_{\text{ring}} = 2287.0$ km, and bulk density $\rho_{\text{bulk}} = 1885$ kg/m³ (Ortiz et al. 2017) with $R^2 = 0.9998$.